

Experiment No. : 01

Name of the Experiment : Verification of Ohm's Law.

Objective:

To verify the following two equivalent forms of Ohm's Law:

- Express I as a function of V and R .
- Express V as a function of I and R .

Theory:

Ohm's law describes mathematically how voltage ' V ', current ' I ' and resistance ' R ' in a circuit are related. According to this law:

"The current in a circuit is directly proportional to the applied voltage and inversely proportional to the circuit resistance".

Formula for voltage:

For a constant value of R , V is directly proportional to I
i.e. $V = IR$

Formula for current:

For a constant value of V , I is inversely proportional to R
i.e. $I = V/R$

Equipments:

- Variable DC power supply - 1 piece
- Digital multimeter (DMM)/ Analog multimeter - 1 piece.
- Resistances: 1 K Ω , 2.2 K Ω , 3.5 K Ω , 5.5 K Ω , 6.9 K Ω , 8.6 K Ω - 1 piece each.
- Trainer Board
- Connecting Wires.

CIRCUIT DIAGRAM:

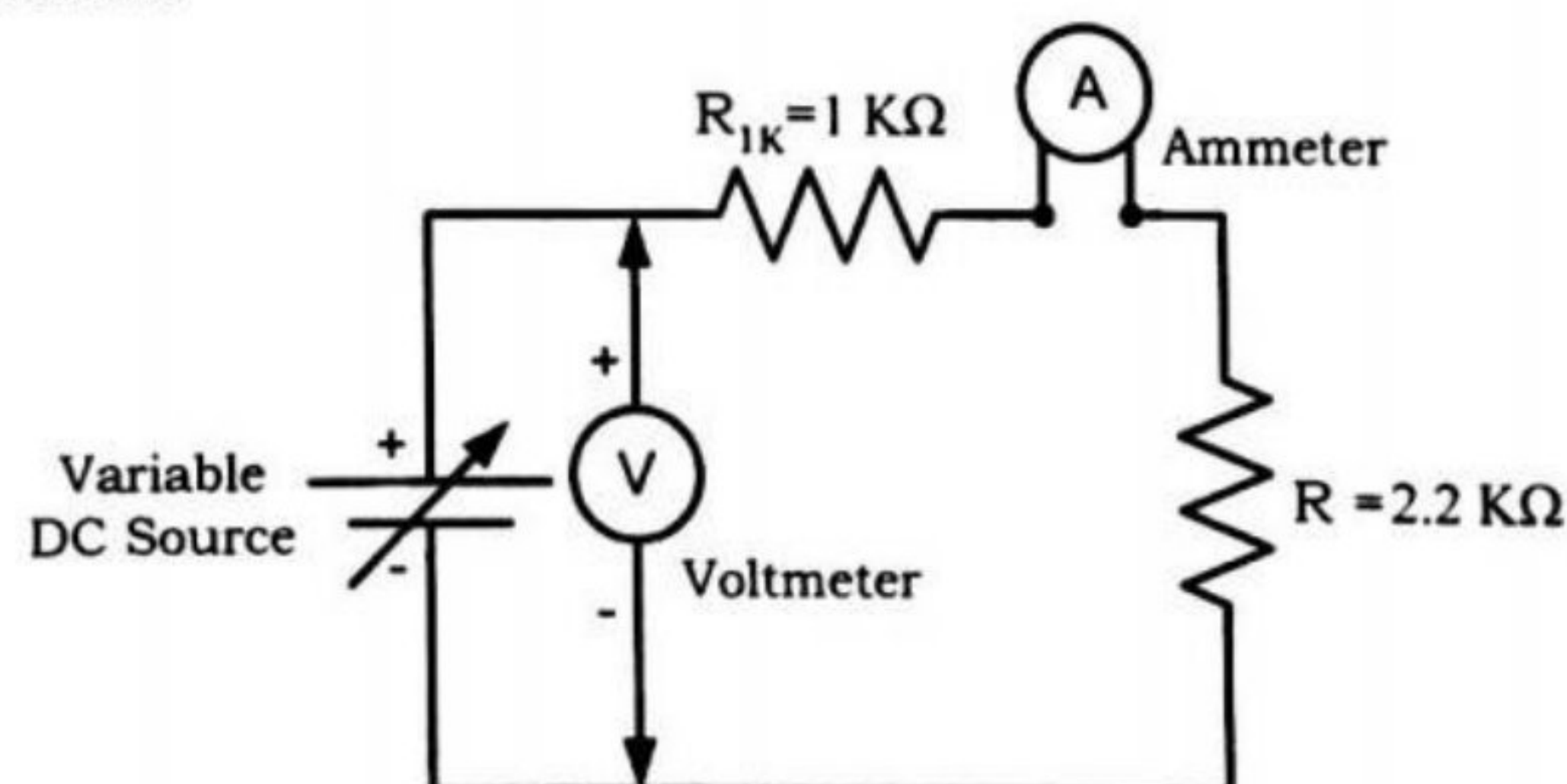


Figure 2.1: Verification of Ohm's Law

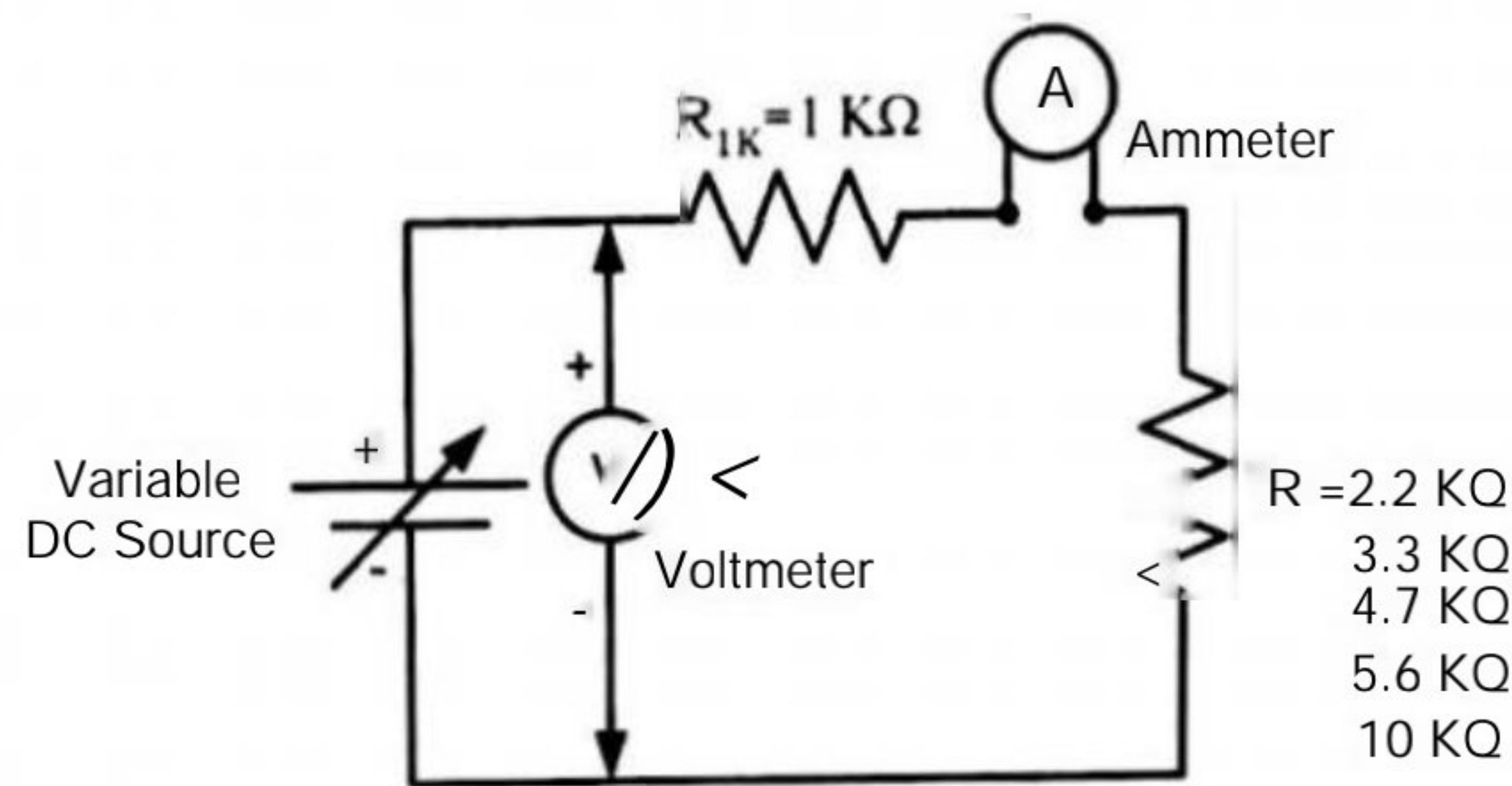


Figure 2.2: Verification of Ohm's Law

Procedures:

Current versus voltage:

- Construct the circuit of Figure 2.1. Do not switch on the power supply.
- Turn on the power supply and adjust it to 500V by using Voltmeter. Measure the current I by ammeter and record it in the Table 2.2.
- Increase the values of voltage as shown in the Table 2.2. Measure the current I in turn and record the values in Table 2.2.
- Calculate the values of current I by using $I = V/R_t$. Use measured values of resistances.

Current versus resistance:

- Construct the circuit of Figure 2.2. Do not switch on the power supply.
- Turn on the power supply and adjust it to 500V by using Voltmeter. Measure the current I by ammeter for $R = 2.2 \text{ K}\Omega$ (Use measured values) and record it in the Table 2.3.
- Turn off the power supply and remove the resistance $2.2 \text{ K}\Omega$. Replace it by resistor $3.5 \text{ K}\Omega$.
- Now turn on the power supply. Measure and record the current I in turn, at each of the resistance settings shown in the Figure 2.2.
- Calculate the values of resistance R_T by using $R_t = V/I$. Use measured values of voltage and current.

Data Sheet:

Table 2.1: Measuring Resistances by using Ohmmeter

Nominal values of R (KS1)	Measured values of R(KII) by using Ohmmeter
1	
2.2	
3.3	
4.7	
5.6	
10	

Table 2.2: Current versus voltage

Supply Voltage (V)	Measured I by using Ammeter (A)	$R_t = R_k + R_{lk}$ [Use measured values of R]	Calculate I (A) $I = V/R_t$
500	0.16	3200	0.16
550	0.17	3200	0.17
600	0.19	3200	0.19
650	0.20	3200	0.20
700	0.22	3200	0.22

Table 2.3: Current versus resistance

Supply Voltage (V)	Measured I by using Ammeter (A)	R_t (K Ω) Use measured values of R	Calculate $R_t = V/I$ (K Ω)
500	0.16	$R_t = R_{1k} + R_{2.2K}$ $R_t = 1000 + 2200 = 3200$	3125
500	0.11	$R_t = R_{1k} + R_{3.5K}$ $R_t = 1000 + 3500 = 4500$	4545.45
500	0.08	$R_t = R_{1k} + R_{5.5K}$ $R_t = 1000 + 5500 = 6500$	6250
500	0.06	$R_t = R_{1k} + R_{6.9K}$ $R_t = 1000 + 6900 = 7900$	8333.34
500	0.05	$R_t = R_{1k} + R_{8.6k}$ $R_t = 1000 + 8600 = 9600$	10000

Signature of the Teacher

Physics
Lab Report # -01

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1 NO; Ans;

We know that,

$$V = IR$$

V = Voltage

I = Current

R = Resistance

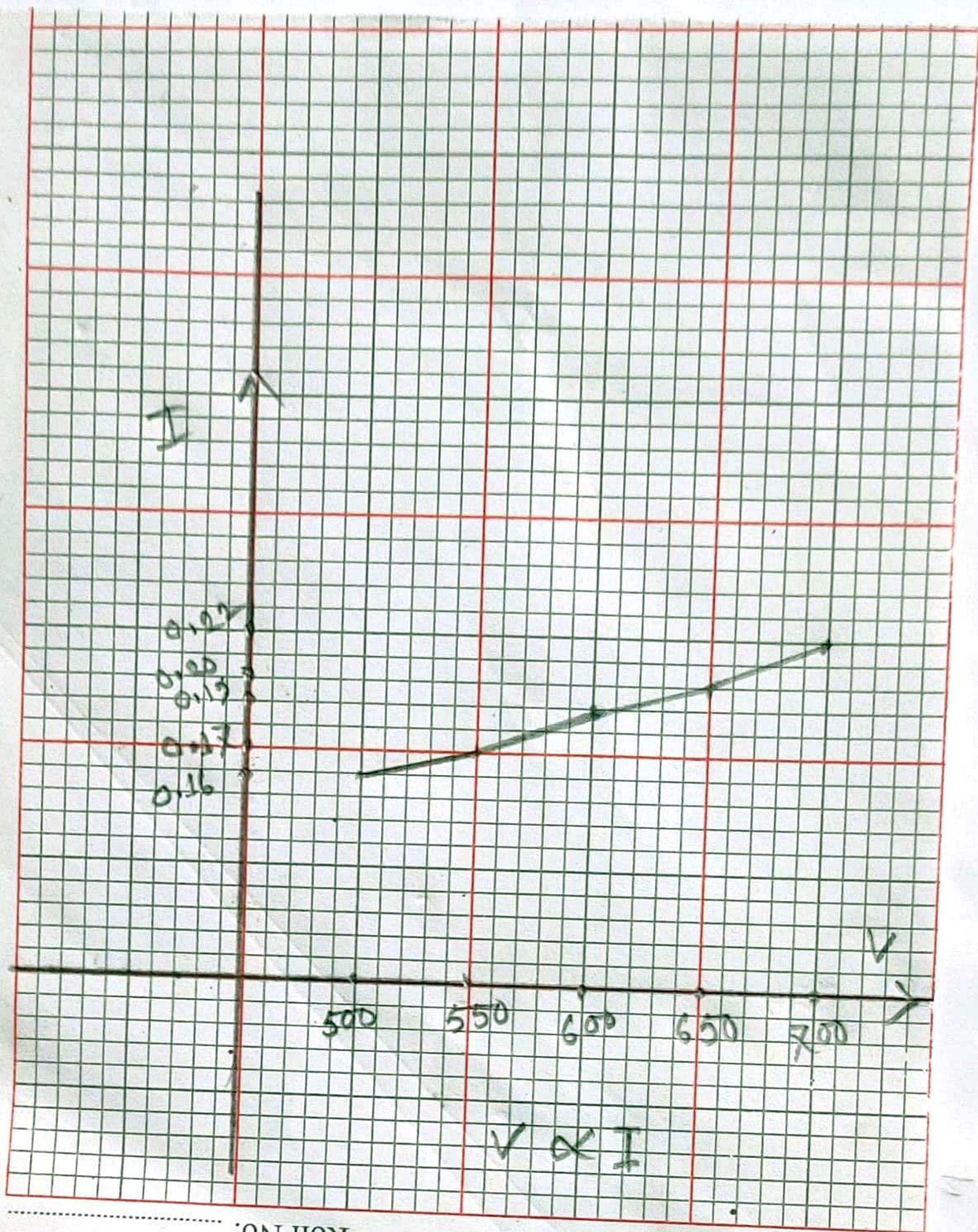
If resistance is fixed

$$V \propto I$$

that means, if voltage is increases then
Current will also increase or if voltage
in decrease current will also decrease

~~2 NO; Ans;~~

2 NO: Ans:

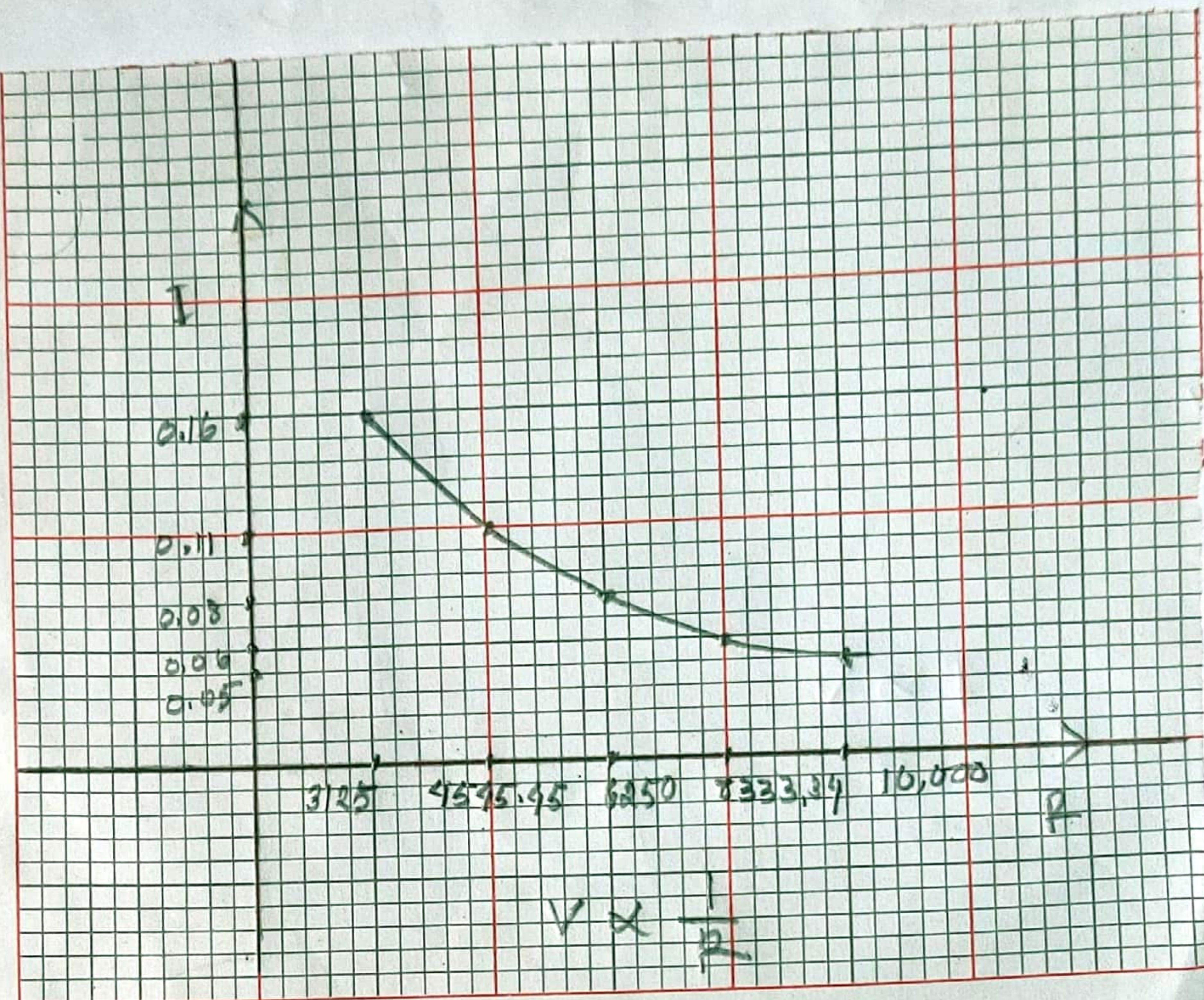


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The graph indicates that, if V increased then current (i) will be increased also.

is

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$$V = IR$$

Here, V is constant

$$\text{So, } I = \frac{V}{R}$$

$$I \propto \frac{1}{R}$$

From the graph we see that, the graph is decrease.